3.4.3 Expansion of $(x+y)^3$ and $(x-y)^3$

$$(x+a)(x+b)(x+c) \equiv x^3+(a+b+c)x^2+(ab+bc+ca)x+abc$$

substituting $a=b=c=y$ in the identity

we get, $(x+y)(x+y)(x+y) = x^3+(y+y+y)x^2+(yy+yy+yy)x+yyy$

$$= x^3+(3y)x^2+(3y^2)x+y^3$$

Thus, $(x+y)^3 \equiv x^3+3x^2y+3xy^2+y^3$ (or) $(x+y)^3 \equiv x^3+y^3+3xy(x+y)$

by replacing y by $-y$, we get

$$(x-y)^3 \equiv x^3-3x^2y+3xy^2-y^3 \text{ (or) } (x-y)^3 \equiv x^3-y^3-3xy(x-y)$$

Example 3.21 Expand $(5a-3b)^3$

Solution

We know that, $(x-y)^3 = x^3-3x^2y+3xy^2-y^3$

$$(5a-3b)^3 = (5a)^3-3(5a)^2(3b)+3(5a)(3b)^2-(3b)^3$$

$$= 125a^3-3(25a^2)(3b)+3(5a)(9b^2)-(3b)^3$$

$$= 125a^3-225a^2b+135ab^2-27b^3$$

The following identity is also used:

$$x^3+y^3+z^3-3xyz \equiv (x+y+z)(x^2+y^2+z^2-xy-yz-zx)$$

We can check this by performing the multiplication on the right hand side.

Note

(i) If $(x+y+z)=0$ then $x^3+y^3+z^3=3xyz$

Some identities involving sum, difference and product are stated without proof

(i) $x^3+y^3 \equiv (x+y)^3-3xy(x+y)$ (ii) $x^3-y^3 \equiv (x-y)^3+3xy(x-y)$

Example 3.22 Find the product of

$(2x+3y+4z)(4x^2+9y^2+16z^2-6xy-12yz-8zx)$

Solution

We know that, $(a+b+c)(a^2+b^2+c^2-ab-bc-ca) = a^3+b^3+c^3-3abc$

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